PHASE 2 | WEEKS 4-6

Segmentation Model Training & Evaluation

Brain CT Hemorrhage Segmentation with Active Learning

1. Executive summary

Phase 2 established and trained three segmentation baselines for multi-class intracranial hemorrhage segmentation: DeepLabV3+ with a ResNet34 encoder, nnU-Net v2 in its 2D configuration, and a compact custom 2.5D U-Net. The models use six output classes (background plus IPH, SAH, EDH, SDH and IVH) and nine-channel inputs constructed from adjacent CT slices and three clinical window views where documented.

nnU-Net v2 has the strongest recorded foreground macro Dice (0.5399) and macro IoU (0.3907). This is a preliminary ranking because validation cohorts, split definitions, checkpoint selection and metric reporting differ. DeepLabV3+ remains a useful comparison model but reports zero Dice for EDH. The custom U-Net is lightweight, although its five foreground output labels are not mapped to clinical class names in the available metrics.

Phase 2 results at a glance

Baseline	Best reported foreground macro Dice	Macro IoU	Validation cohort in report	Best point
nnU-Net v2 (2D, fold 0)	0.5399	0.3907	33 cases	Native best checkpoint; 50-epoch target
DeepLabV3+ (2.5D, ResNet34)	0.4534	0.3347	28 volumes / 924 slices	Epoch 19; validation loss 0.5601
Custom 2.5D U-Net	0.4256	Not reported	39 studies	Epoch 33; validation loss 0.8055

PHASE 2 / SETUP

2. Pipeline and model configurations

2.1 Shared segmentation task

- Input data: 3D NIfTI CT volumes and voxel-level masks are documented for the DeepLabV3+ pipeline; the U-Net and nnU-Net reports describe reusing the prepared nine-channel dataset representation.
- 2.5D input for DeepLabV3+ and custom U-Net: slices z-1, z and z+1 are stacked. Each slice contributes brain, subdural and bone window channels, giving 9 channels at 512 x 512 resolution.
- Clinical windows: Brain WL 40 / WW 80; Subdural WL 75 / WW 215; Bone WL 400 / WW 1800.
- Output: six-class segmentation logits for background, IPH, SAH, EDH, SDH and IVH. DeepLabV3+ documentation specifies the mapping from BHSD raw labels into this target ordering.
- Training safeguards documented in the DeepLabV3+ run include patient-level partitioning, class-aware slice sampling, weighted Cross-Entropy plus multiclass Soft Dice, checkpointing, mixed precision, gradient clipping and early stopping.

2.2 Baselines documented

Item	DeepLabV3+	nnU-Net v2	Custom 2.5D U-Net
Model form	2.5D semantic segmentation	2D configuration, fold 0	2.5D U-Net
Input / output	9 channels / 6 classes	9 CT-derived channels / 6 classes	9 x 512 x 512 / 6 output maps
Backbone / size	ResNet34, ImageNet pretrained	Native nnU-Net configuration	483,686 trainable parameters
Dropout evidence	Decoder dropout 0.2; notebook checks for dropout presence	Report does not document dropout level or MC Dropout verification	4 Dropout2d layers, p = 0.2
Optimization	AdamW; LR 3e-4; cosine schedule; weighted CE + Soft Dice	Native nnU-Net training setup; 50-epoch target	AdamW; LR 0.001; cosine schedule; CE + foreground Soft Dice

2.3 Data partitions reported

Model / report	Training	Validation	Development / other detail
DeepLabV3+	134 volumes; 3,502 executed slices	28 volumes; 924 manifest slices	30 development volumes; split manifests list 4,503 train, 924 val and 977 development slices
nnU-Net v2	Not stated as a training-case count in report	33 validation cases	Dataset501_BrainCTICH25D; 2D configuration; fold 0
Custom U-Net	153 studies	39 studies	20% validation fraction; seed 42

PHASE 2 / TRAINING

3. Training outcomes and overall comparison

Model performance is summarized using foreground macro metrics so the dominant background class does not control the headline score.

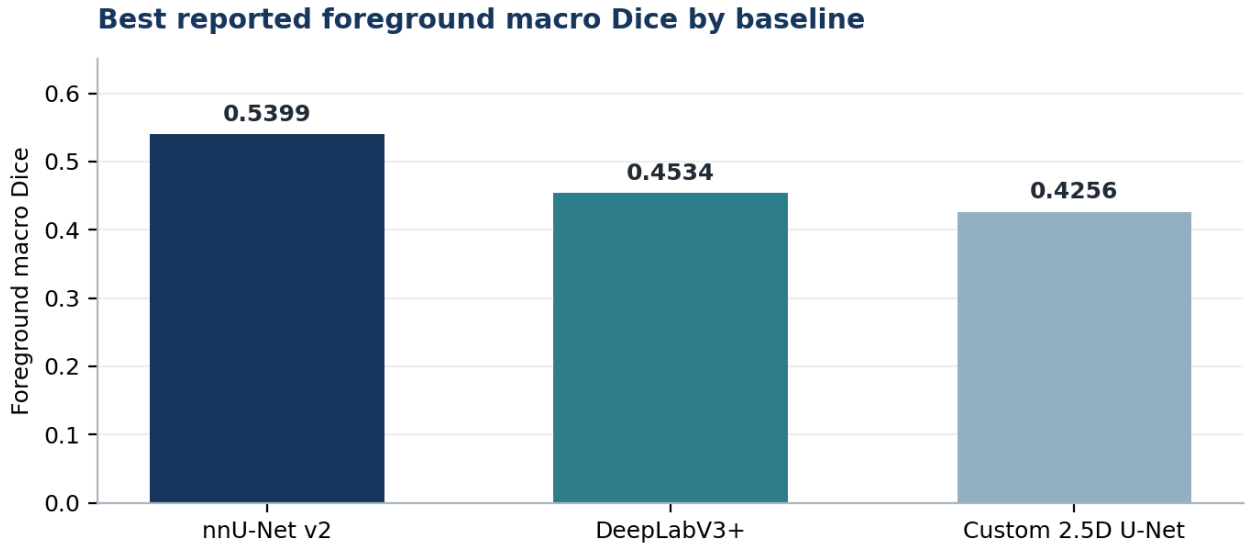


Figure 1. Best reported foreground macro Dice in the three supplied baseline reports. Validation cohorts and evaluation conditions differ; this chart is descriptive, not a controlled benchmark.

3.1 Reported training checkpoints

Baseline	Training record	Reported selected result	Training signal / observation
DeepLabV3+	31 epochs completed; stopped by early stopping	Best at epoch 19; macro Dice 0.4534; macro IoU 0.3347; val loss 0.5601	Training loss continued to fall after the best validation point, while validation performance fluctuated and declined.
nnU-Net v2	50-epoch target; fold 0	Macro Dice 0.5399; macro IoU 0.3907; native best checkpoint	Per-class Dice is non-zero for all five named hemorrhage classes in the report.
Custom 2.5D U-Net	35 epochs in saved history	Best recorded macro Dice 0.4256 at epoch 33; val loss 0.8055	Training loss fell from 2.0219 at epoch 1 to 0.6241 at epoch 35.

3.2 Interpretation

- nnU-Net v2 is the current leading candidate by its own reported foreground macro Dice and macro IoU, with reported Dice above zero for each named hemorrhage class.
- DeepLabV3+ performs strongly on IPH and IVH, but its EDH output failed on the reported validation cohort. Its best validation macro Dice was reached at epoch 19; the later loss trend is consistent with overfitting.
- The custom U-Net offers a compact, 483,686-parameter baseline. Its strongest exported class is Label 2 (Dice 0.7596), but the clinical identity of Labels 1-5 is not established in the report.

PHASE 2 / CLINICAL CLASSES

4. Class-wise validation results

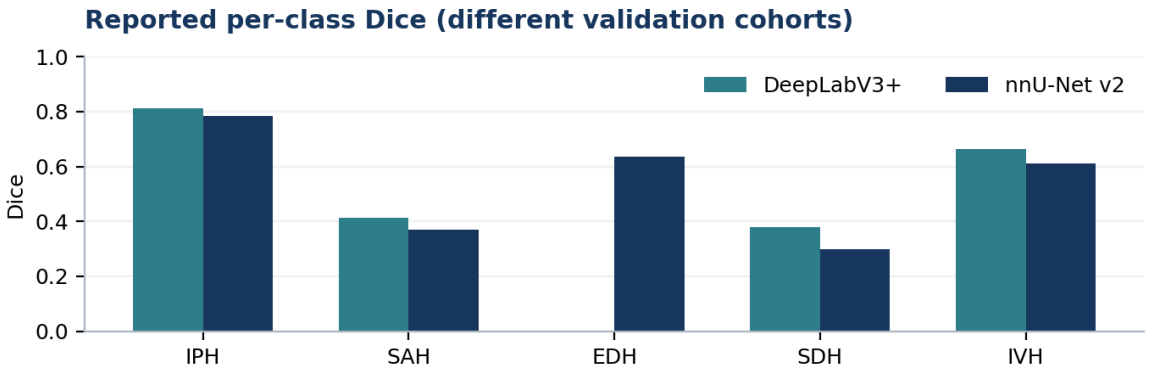


Figure 2. Reported Dice for the two models with clinically named output classes. The validation cohorts differ, so the chart should not be read as a same-case comparison.

4.1 DeepLabV3+ validation metrics

Class	Dice	IoU	Precision	Recall
Background	0.9991	0.9982	0.9991	0.9992
IPH	0.8115	0.6828	0.7892	0.8351
SAH	0.4136	0.2607	0.3875	0.4435
EDH	0.0000	0.0000	0.0000	0.0000
SDH	0.3790	0.2338	0.4984	0.3058
IVH	0.6630	0.4959	0.6689	0.6572
Foreground macro	0.4534	0.3347	Not reported	Not reported

4.2 nnU-Net v2 validation metrics

Class	Dice	IoU	Precision	Recall
Background	0.9995	0.9990	0.9994	0.9996
IPH	0.7836	0.6441	0.7689	0.7988
SAH	0.3691	0.2264	0.4084	0.3368
EDH	0.6367	0.4670	0.6419	0.6316
SDH	0.2986	0.1755	0.3136	0.2849
IVH	0.6117	0.4406	0.7548	0.5143
Foreground macro	0.5399	0.3907	Not reported	Not reported

4.3 Class-level takeaways

- IPH is the strongest class for both clinically named reports: Dice 0.8115 for DeepLabV3+ and 0.7836 for nnU-Net v2.
- SAH and SDH remain difficult for both models, with Dice below 0.42 and below 0.38, respectively, across the reported runs.
- EDH is the largest divergence: DeepLabV3+ reports Dice 0.0000, while nnU-Net v2 reports 0.6367. The comparison report recommends checking class distribution, label mapping, sampling and per-case predictions.

PHASE 2 / ADDITIONAL BASELINE

5. Custom 2.5D U-Net and saved evidence

5.1 Recorded configuration

Parameter	Recorded value
Architecture / parameters	Custom 2.5D U-Net; 483,686 trainable parameters
Input / output	9 x 512 x 512 input; six output channels
Dropout	Four Dropout2d layers, p = 0.2
Loss / optimizer	Cross-Entropy + foreground multiclass Soft Dice; AdamW
Learning rate / weight decay	Initial LR 0.001; weight decay 0.0001
Schedule / precision	CosineAnnealingLR; bfloat16 mixed precision
Training run	35 epochs in saved history; batch size 64
Validation split	153 training studies / 39 validation studies; seed 42

5.2 Per-label validation results

Exported label	Dice	Ground-truth voxels	Predicted voxels	True positives
Label 1	0.2903	29,268	43,739	10,597
Label 2	0.7596	493,253	418,258	346,177
Label 3	0.4778	106,742	127,036	55,853
Label 4	0.2782	137,311	137,393	38,214
Label 5	0.3243	153,832	72,911	36,763
Foreground macro Dice	0.4256	Not applicable	Not applicable	Not applicable

5.3 Evidence and artifacts described in reports

- DeepLabV3+: best and latest checkpoints, epoch-level training history, validation metric exports, training curves, confusion matrix, dataset inspection and prediction overlays are shown or listed in the pipeline report.
- nnU-Net v2: native training progress, a normalized validation confusion matrix, per-class metric chart and per-class Dice versus epoch plot are included in the report.
- Custom U-Net: training configuration and model card, saved training history, validation per-class metrics, confusion matrix, split manifest and best/last model checkpoints are listed as artifacts.

Metadata issue to resolve: the custom U-Net report contains 35 epochs in the saved history, while the comparison report notes that the model card separately describes the run as a three-epoch smoke test. The final run record should be reconciled before claiming full reproducibility.

PHASE 2 / REVIEW & HANDOFF

6. Evaluation quality, limitations and next steps

6.1 What Phase 2 established

- Three training baselines have recorded validation results: DeepLabV3+, nnU-Net v2 and custom 2.5D U-Net.
- DeepLabV3+ reports a complete named-class table covering Dice, IoU, precision and recall; nnU-Net v2 also reports all four metrics by named class.
- Training-history visualizations, confusion matrices, per-class results and DeepLabV3+ prediction overlays are available for review.
- Dropout is documented for DeepLabV3+ and the custom U-Net. MC Dropout uncertainty scoring and active-learning selection remain future implementation tasks.

6.2 Limitations that affect interpretation

Issue	Why it matters	Required resolution
Different validation cohorts	A higher reported score may reflect cohort composition as well as model quality.	Evaluate saved checkpoints on one fixed, patient-disjoint cohort with a shared metric implementation.
Custom U-Net label names	Label 1-5 cannot be safely matched to clinical classes.	Verify the mask ID mapping and update metrics/model card before class-wise comparison.
DeepLabV3+ EDH Dice = 0	The model did not segment EDH in the reported validation run.	Inspect EDH support, sampling frequency, mask mapping, false negatives and per-case overlays.
nnU-Net augmentation audit	Inputs are already windowed channels; source inspection does not prove HU-space jitter was active.	Do not claim +/-50 HU augmentation unless it is applied to raw-HU images and verified in the actual training path.
nnU-Net dropout evidence	The Week 5 report does not state the dropout level or confirm stochastic prediction variation.	Confirm dropout configuration and non-zero MC prediction variance before reusing it in the uncertainty stage.
Inference workflow coverage	The reports show prediction/validation artifacts, but do not establish one shared, verified inference script for all models.	Document and test the end-to-end input-to-mask workflow for the selected model.

6.3 Handoff to Phase 3: failure analysis

- Freeze the study IDs for the common comparison set and confirm that patient-level leakage is absent.
- Recompute per-class Dice, IoU, precision, recall and confusion matrices from saved predictions using one metric implementation. Explicitly exclude background from foreground macro averages.
- Review FP and FN patterns per class, starting with DeepLabV3+ EDH failure and the lower reported SAH/SDH scores. Record visual evidence beside the numeric metrics.
- Reconcile the custom U-Net run metadata and label mapping. Document nnU-Net augmentation and dropout behavior from the executed pipeline, not only from source-level configuration.
- Preserve the Phase 2 score table as the reported baseline, but replace it with common-split metrics before making a final model-selection claim or any active-learning label-efficiency claim.